

Abandoned leads

This file details every lead we explored during development and testing that was finally abandoned and explains the reasons why.

I – Classification by patch

As we believed the ESP32-S3 Dev Module to be not powerful enough to infer object detection models, we first used a system where the image was first cut into different patches and then a classification model was inferring on each of those patches. This method provided good results. However, the main challenge was to find a way to only have a single insect in each patch. Although this could have worked, when we realized that the ESP32-S3 could run Object Detection models we abandoned this idea as it would have been pretty complex to set up.

II – Object Detection Models

1- YOLO models

The YOLO (You Only Look Once) models are a reference in the Object Detection model world. They are very easy to train, using the Ultralytics Hub platform, that provides a friendly user interface. Training a YOLO model using this platform doesn't require writing a single line of code. Although YOLO models showed great accuracy using the raw Pest24 dataset, every iteration I tried resulted in a model that was way too heavy for the ESP32 (Minimum was 2.5MB tflite model, using nano versions).

2- ROBOFLOW models

Similar to Ultralytics Hub, ROBOFLOW offers a training platform for object recognition models making it very easy to train models. The models offered by this platform are SSD-MobilenetV2 based. But because you are not allowed to customize the model and can only choose from presets, I was unable to get a model that was light enough to fit in the ESP32.

3- PicoDet models

I explored the idea of using a PicoDet model as these are supposed to be the amongst the lightest available, however, as they need to be developed using the PaddlePaddle framework instead of either TensorFlow or Pytorch, I ultimately didn't train one. Additionally, most of the resources about these models and the required tools are only available in Chinese. Finally, there is no known way to export PicoDet models to the .tflite format as those models are originally designed for mobile phones.

4- Edge Impulse models

Edge Impulse models are pretty much the only type of IA object detection models that come up for ESP32 S3 Dev Module when searching online. The training platform is designed to work with this specific microcontroller which makes it seem like a very good choice. However, it is not designed for tasks as complex as insect detection, allowing only 1 hour of free training. A typical setup to use this platform is ~200 manually annotated images and 90x90 image resolution, definitely not 25 000+ images and a >=240x240 image resolution.

5- NanoDet models

NanoDet models are PicoDet like models but use the TensorFlow framework. There are not many resources available online (pretty much only the official GitHub repository). I got some okay results when training NanoDet models that fit the ESP32-S3 Memory requirements. When training these models, make sure to use a starting point (available on GitHub), and reach the desired weight by pruning (supervised learning) if needed. However, the model couldn't ultimately be used on ESP32 probably because it was too complex for the MicroTflite Arduino library (too many Transform operations would be my guess). Only output zeros as predictions, even when the preprocessing is correct.

III- RBG565 camera format

Although it seems weird to capture the image in the JPEG format and then convert it to RGB format for image inference instead of directly capturing in RGB format, the reason is that RGB capture causes a crash due to memory overflow. As soon as the capture format was switched to RGB, the camera capture kept failing without me finding any solution.

IV- Use of SD card

Using a SD to store images or data was an option during development, however the ESP32-S3 never detected the SD card, even when I tried by brute force every single GPIO combination possible.